Submission Cover Page

Exercise number: 1

Group code: SMNGRP05

Self-evaluation score (0-100): 90

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GitHub repository: https://github.com/reem-aw/SignalMemoryNet

Late submission: no

SignalMemoryNet — HW1 Lab Report

1. Problem Setup

We generate sine signals at four discrete frequencies (2, 5, 10, 20 Hz), sampled at 100 Hz for 10 seconds, with random amplitude in [0.8, 1.2], random phase, and additive Gaussian noise at three levels (σ = 5 %, 10 %, 20 % of amplitude). The task is two-fold per 10-sample window: (i) classify the underlying frequency (4 classes), (ii) reconstruct the clean window (denoising). A shared backbone feeds a multi-task head producing 4 logits (cross-entropy) + 10 regression outputs (MSE).

2. Architectures

MLP — fully-connected baseline that ignores temporal order: FC(10→64)–ReLU–FC(64→64)–ReLU–Head(14).

RNN — single-layer tanh nn.RNN with hidden size 32; the final hidden state h_T is fed to the same head.

LSTM — single-layer nn.LSTM with hidden size 32; gating (input, forget, output) mitigates the vanishing-gradient issue and is theoretically expected to improve denoising under high noise and longer temporal dependencies.

Although LSTM is theoretically better suited for noisy sequential data, the MLP achieved the best reconstruction MSE in our setup. This is likely due to the short context window (10 samples) and the large separation between frequencies, making the task relatively easy even without recurrent memory.

3. Training Setup

Optimizer Adam (lr=1e-3), batch size 128, up to 20 epochs, early stopping (patience 5) on the validation loss. Records per frequency: train 30 / val 8 / test 8, sliding windows of length 10 with stride 5. Seed = 42 for full reproducibility.

4. Results

Model	Test Acc	Recon MSE	MSE (σ=5%)	MSE (σ=10%)	MSE (σ=20%)
MLP	0.986	0.0057	0.0011	0.0031	0.0113
RNN	0.978	0.0127	0.0057	0.0093	0.0207
LSTM	0.985	0.0100	0.0051	0.0069	0.0164

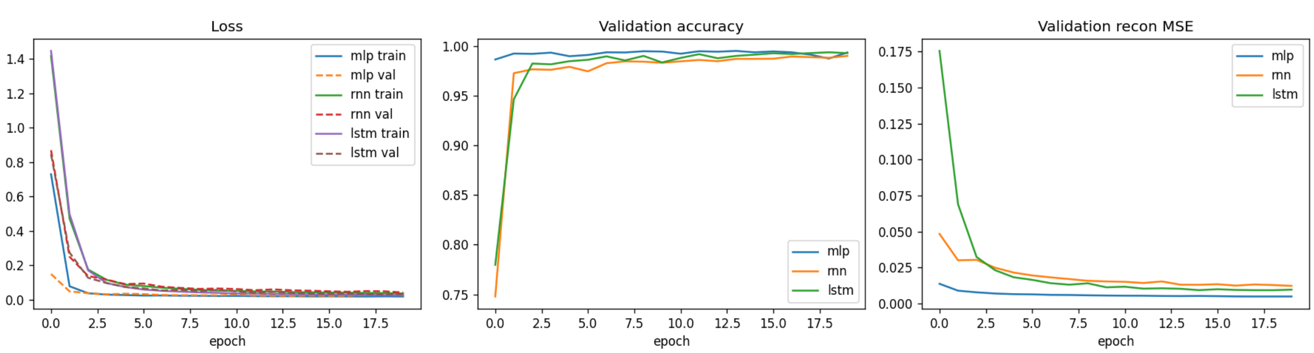

Figure 1 — Training/validation loss curves for MLP, RNN, LSTM.

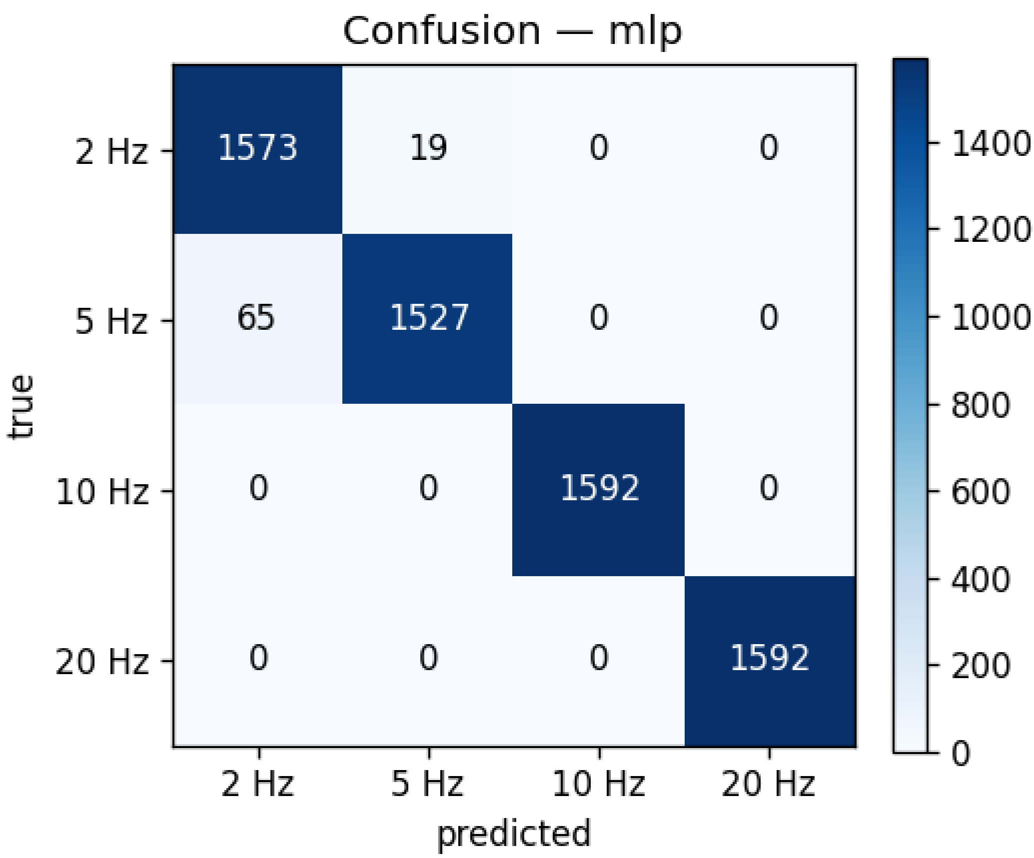

Figure 2 — Confusion matrix (MLP).

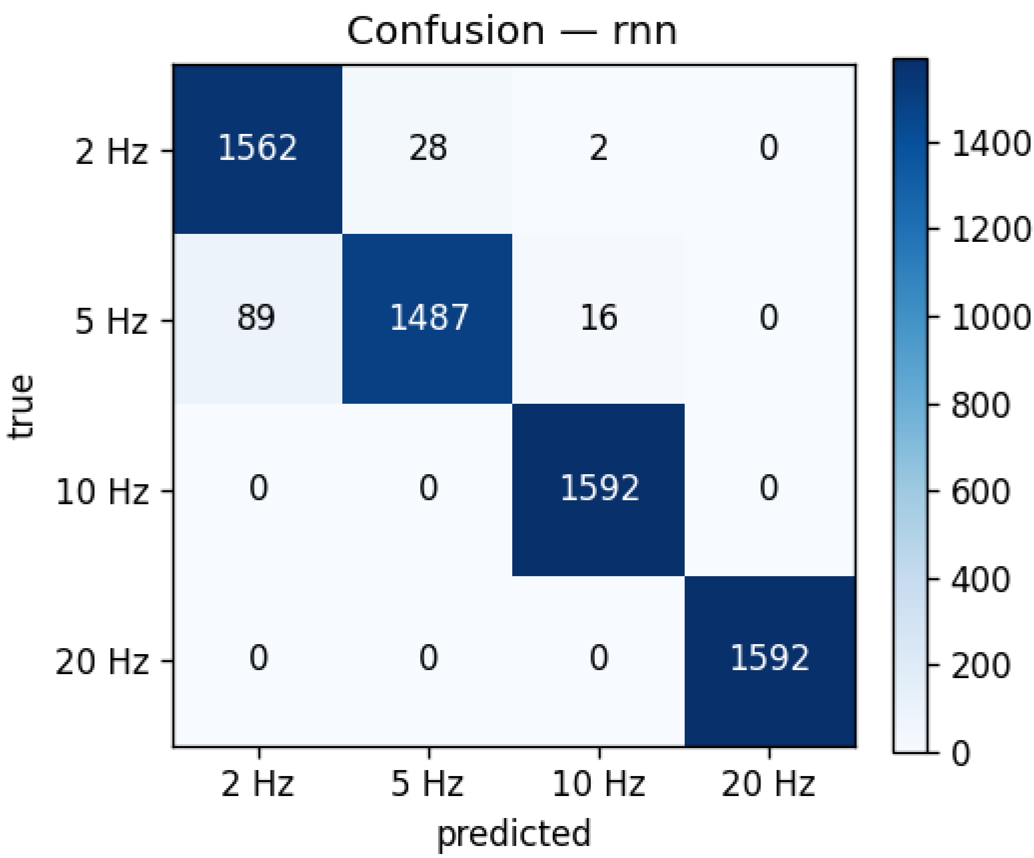

Figure 3 — Confusion matrix (RNN).

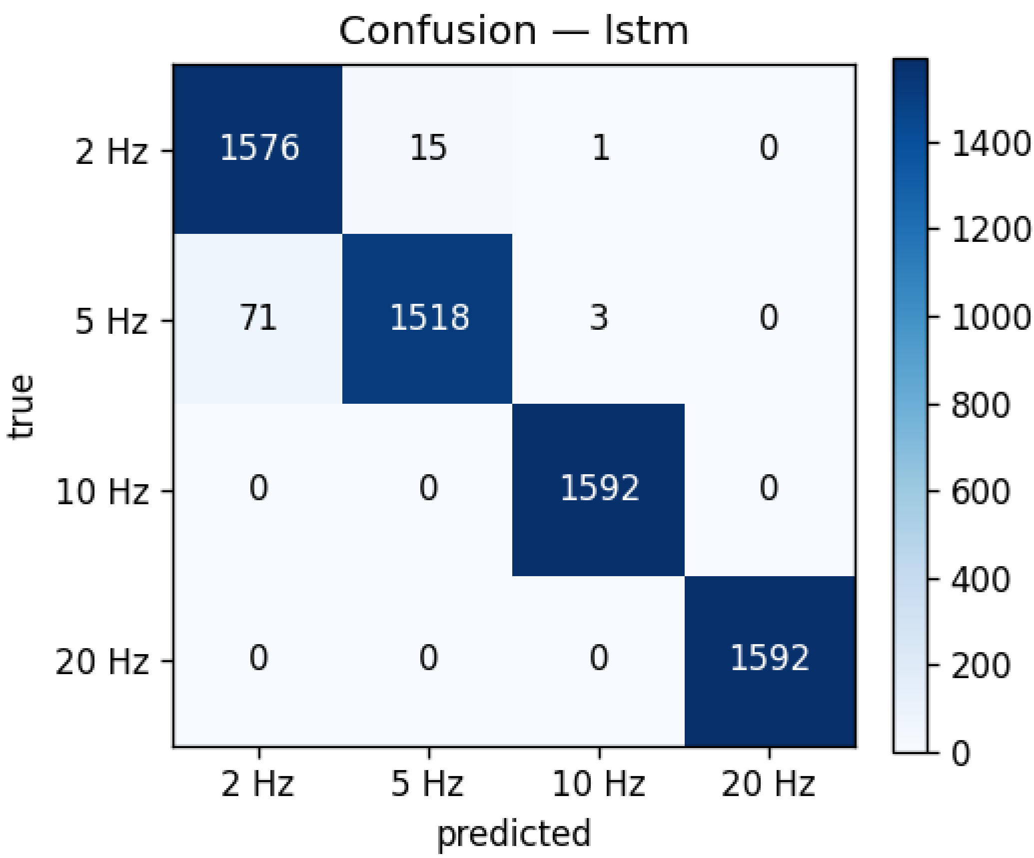

Figure 4 — Confusion matrix (LSTM).

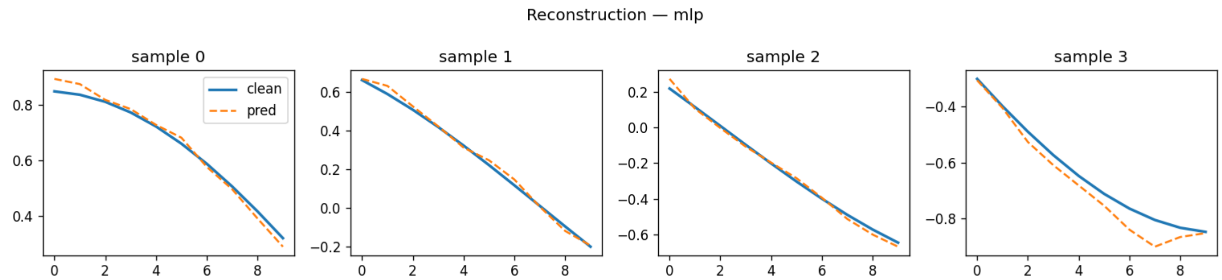

Figure 5 — Reconstruction examples (MLP).

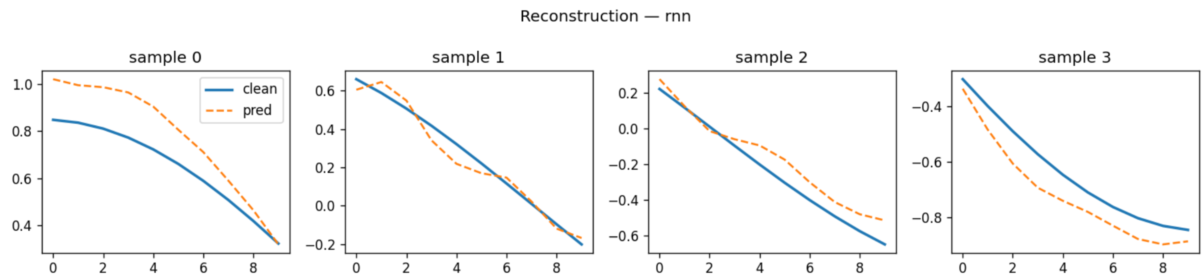

Figure 6 — Reconstruction examples (RNN).

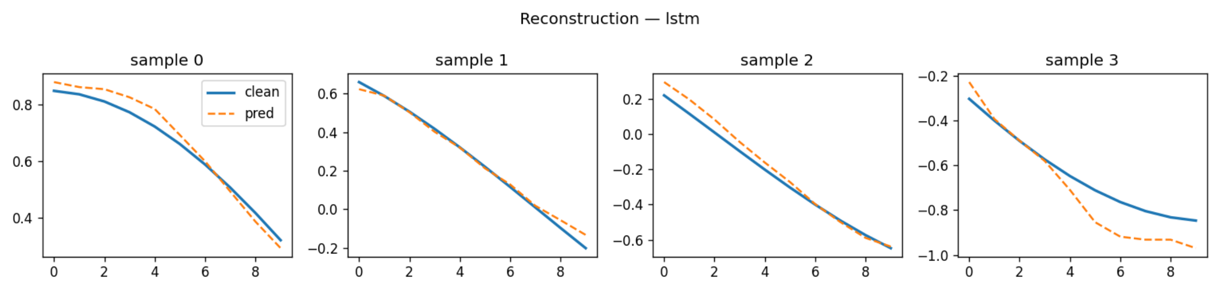

Figure 7 — Reconstruction examples (LSTM).

5. Discussion

All three models achieved more than 97% classification accuracy thanks to the strong frequency cues present even within a short 10-sample window at 100 Hz. While recurrent architectures are theoretically better suited for sequential denoising tasks, the MLP achieved the best reconstruction MSE in our setup. This is likely because the context window was very short and the four frequencies were well separated, allowing the fully connected model to capture the relevant patterns without relying heavily on temporal memory. The LSTM nevertheless showed more stable behavior than the vanilla RNN, especially under higher noise levels.

6. Conclusion

In this work we compared MLP, RNN, and LSTM architectures on a noisy sine-wave reconstruction and classification task. All models achieved high classification accuracy due to the strong spectral separation between frequencies. Surprisingly, the MLP achieved the best reconstruction MSE under the chosen setup, likely because the context window was very short. The results highlight how sequence models become more beneficial as temporal dependencies grow longer and more complex.